EULER - Mathematical editing by voice input for people with visual impairment

Tribeth Rivas-Pérez [1], Abraham Meza-Vega [2], Verónica Mora-Lezcano[3], Dionisio Palacio-Amador[4], Mario Chacón-Rivas[5]

***Abstract*—Access to mathematical content has been limited for the visually impaired population, since some of the technological tools to access information require specific skills, abilities and / or prior knowledge that exclude these people by impeding access to the tool. This paper presents a proposal to improve the accessibility of mathematics for people with visual disabilities, based on the use of information and communication technologies, that allows students to advance at their own learning style. This applied research in process has allowed us to identify alternatives to make mathematics accessible according to the needs of the population with visual disabilities, taking into account that the development is being carried out in a living lab of people with visual disabilities, so it is presented the implemented model of a voice input component in the mathematical editor called EULER, as an alternative to entering basic and compound mathematical expressions, which in turn are accessible in reading and editing. A learning model based on neural networks is used, for which voices of 145 volunteers were used for training. Currently three potential users have evaluated the development of the voice component in a functional way. As a result, opportunities for improvement are obtained in terms of accuracy and proximity when interpreting verbal commands. This research allows us to identify alternatives to make mathematics accessible.**

***Keywords*—Voice input, Human Computer Interaction, Mathematical Editor, Visual Disability, Neural networks.**

I. Introduction

In the process of inclusion of people with disabilities, support products have been developed that facilitate their autonomy, which can be any type of product (including devices, equipment, instruments and software) specially manufactured or available in the market for people with disabilities [1]. This paper refers to a support product as a software component for people with visual disabilities.

The inclusion of Information and Communication Technologies (ICTs) as a support product in education has favored the elimination of barriers to access information, because in this area if the student wishes, he can move at his own pace and learning style [2].

Regarding education, in Costa Rica Law 7600 in its chapter I, article 14 establishes that access to education must be guaranteed to all persons, regardless of their disability [3]. However, during the investigation and work with people with visual disabilities there have been deficiencies, specifically in the mathematics subjects.

On the other hand, the United Nations Organization (UN), in various communiques, has urged member states to recognize and promote the principle of equal opportunity for education at the primary, secondary and higher levels for children, youth and adults with disabilities in integrated environments [4].

The use of technological tools represents an alternative to provide adequate school material to the possibilities of each student [2], therefore, it is recommended for people with visual disabilities the conversion of text to voice and the recognition and digital recording of voice, in order to facilitate the entry and exit of information from technological devices.

In the case of mathematics, which is a pillar in current education, manipulate tools through voice, and that they understand and represent mathematical expressions, it can mean an important advance to make mathematics accessible which is one of the main objectives from EULER.

Within the literature there are several works that focus on supporting the visually impaired user for reading and representing mathematics [5]- [6], however, these systems have the disadvantage that they are not focused on people with visual disabilities, therefore, they omit aspect such as audio confirmation and / or only incorporate basic mathematics.

Consequently, in this work we focus on incorporating functionality through voice to a tool under development that allows ease of use, regarding the interaction and feedback between the user and the system (EULER).

The advantage of this system over similar works is that it offers the visually impaired user the option of creating verbally constructed mathematical expressions, which are represented as if they had been entered using the keyboard.

II. Tools that allow mathematical editing through voice input

Regarding digital tools that incorporate mathematical text input through voice, a literature review was made, as a result, Math Talk, VISO Calculator, EquatIO, Mathifier, CamMath

[1]T. Rivas-Pérez, is with Inclutec, Instituto Tecnológico de Costa Rica, Cartago, Costa Rica, (e-mail: jorivas@ic-itcr.ac.cr).

[2]A. Meza-Vega, is with Inclutec, Instituto Tecnológico de Costa Rica, Cartago, Costa Rica, (e-mail: ameza@ic-itcr.ac.cr).

[3]V. Mora-Lezcano, is with Inclutec, Instituto Tecnológico de Costa Rica, Cartago, Costa Rica, (e-mail: vimora@tec.ac.cr).

[4]D. Palacio-Amador, is with Inclutec, Instituto Tecnológico de Costa Rica, Cartago, Costa Rica, (e-mail: dpalacio@ic-itcr.ac.cr).

[5]M. Chacón-Rivas, is with Inclutec, Instituto Tecnológico de Costa Rica, Cartago, Costa Rica, (e-mail: machacon@itcr.ac.cr). ORCID: 0000-0002-8857-162X

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and TalkMaths have been identified as digital tools designed to facilitate the editing of mathematical expressions.

Here are some features of these tools:

- MathTalk: Voice command dictation and use of lexical indicators for disambiguation [5].
- Viso Calculator: Scientific calculator VISO (Voice Input Speech Output), mouse interaction, keyboard and voice shortcuts, allows zooming, high contrast and narrating the expression [7], [8].
- Cam Math: Entering expressions by voice [9].
- TalkMath: Management of mathematical documents by voice, supports mathematical simbology for intermediate secondary level, for example, calculation and matrices, export expressions to MathML or LaTeX and detection of syntactic errors [10], [11].
- Mathifier: Edit mathematical formulas by voice input, recognize the expression and convert to LaTeX. It is a prototype and does not specify the population [12], [13].
- EquatIO: Entering expressions by keyboard, handwriting or voice, inserting expressions in Google Docs, Word, Google Sheets, Slides, Forms and Drawing [6].

A summary of the characteristics of each of the related works is shown in Table 1, there can be seen that EULER meets all the desirable characteristics in voice interpretation for a mathematical editor, thus achieving a useful system target users.

The characteristics evaluated in the different tools are represented as follows in Table I:

- Voice input = VI.
- Focused on visual disability = VD.
- Makes use of screen reader = SR
- It incorporates calculator = C.
- Perform auditory confirmation = AC.
- It is a mathematical editor = ME.

TABLE I
TOOLS COMPARISON THAT INCORPORATE VOICE INPUT

Toos	VI	VD	SR	C	AC	ME
MathTalk	✓	✓	✓	X	✓	✓
VISO Calculator	✓	✓	✓	✓	X	X
Cam Math	✓	X	✓	X	X	X
TalkMath	✓	X	X	X	X	X
Mathifier	✓	X	X	X	X	X
EquatIO	✓	X	X	X	X	X
EULER	✓	✓	✓	✓	✓	✓

Voice input is an efficient input method for blind people [14]. In the incorporation of voice input of mathematical expressions for EULER, as main characteristics, focused for blind people and low vision, the following are available:

- Voice input of mathematical expressions, as an alternative to using the keyboard.
- EULER handles accessible reading of mathematical expressions with screen readers Non-Visual Desktop Access (NVDA) and Job Access With Speech (JAWS).
- It incorporates a calculator with auditory output.
- Confirmation of the expression that was inserted, this allows people who are blind or have low vision to corroborate if the result of voice entry is as expected.

As part of the development of this component of voice input, it is known the contribution for students who have visual impairment and also have some level of motor disability due to cerebral palsy or other cause.

III. MODEL, INTERPRETATION AND REPRESENTATION OF MATHEMATICAL EXPRESSIONS THROUGH AUDIO FOR EULER

The editing of mathematical expressions through voice input involves the pronunciation of verbal indications (such as preset commands), the interpretation of the indications and the translation and visual representation of the mathematical expression.

For the above, a learning algorithm must be selected to define the prediction model of mathematical symbols and expressions. The task of the learning algorithm is to learn the weights for the model. The weights describe how the model have to behave in relationship to the data [15]. On the other hand, a prediction model is the result obtained from the training phase [16].

A. Learning and training algorithm

To carry out the process of entry, interpretation, translation and representation in EULER, the learning algorithm had to be selected and then the training phase had to be executed to define the prediction model.

1) Learning algorithm selection: Convolutional neural networks work by dividing and modeling information into multiple layers, each of these layers has a higher level of abstraction compared to the previous one, which allows it to discern between basic elements or complex elements [17], [18]. Acoustic models or voice inputs have rapid progress using neural networks [19], [20]. In addition, convolutional neural networks work by converting audio inputs into spectrograms to make the classifications [21]. Based on this, the convolutional network algorithm was selected to be applied in EULER.

For the implementation of this algorithm we used the open source library TensorFlow developed by Google that is used to meet needs such as image recognition, voice, data prediction, decision making, machine translation, among others [22]. This library is selected because it offers robust models with very wide support for different platforms.

2) Training and model creation: Prior to the training stage, a set of data (*dataset*) is required to be supplied to the neural network, which is used to establish the parameters that will be used to make the predictions.

The dataset audios for the training were classified as: numbers, Latin alphabet, delimiters and mathematical symbols. The work is focused on the teaching-learning process of the area of mathematics for people with visual disabilities, so the prediction model that is sought must be mathematical.

For the construction of the dataset, support was obtained from volunteers, 145 people so far, 81 adults (20 years and older),

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64 children (5-12 years), who agreed to record their voices for the dataset. The audios obtained were divided into two parts, 80% for training and 20% for tests and validation. The division is done in this way on the recommendation of TensorFlow [23].

The training audios are taken by the network to try to find patterns in them and generalize future verbal entries that have not been entered, distinguishing from the set of possibilities (moments of silence, nonverbal and background noises and accepted commands) to determine which interpretations are more likely to be the searched word [24].

The validation audios are used to verify that the network is not memorizing the training data, this is a usual process in this type of neural networks. Once the network complies with the training phase, one of the validation audios is supplied to verify that it is learning correctly, in advance the network is given the possibility to verify if it answer was correct with a file which it is created manually and contains the answers, indicating to which set the audio being tested belongs. Additionally, the training program automatically stores this data which the user can consult during the training phase of the network [24], as shown in Figure 1. The graph on the left shows how the network is improving accuracy and on the right it shows how the difference between the predicted result and the correct answer is reduced.

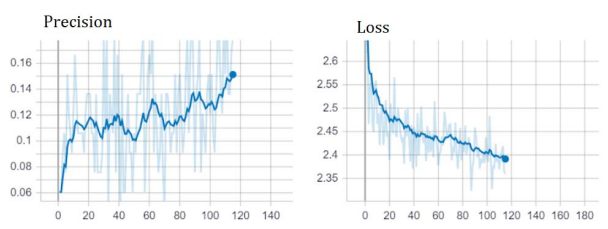


Fig. 1. Convolutional neural network training stats

In addition, to improve the recognition process, the neural network is trained with data that is not relevant for the interpretation of mathematical content, for example, background noises, moments of silence and involuntary errors in speaking, this is usual in this type of networks.

At the end of the training, the TensorFlow library indicated that the neural network has approximately 82% accuracy.

B. Interpretation and representation of the input audio

The first step is to capture the audio, for this, the TensorFlow convolutional neural network, defines a time window in which a user is expected to enter some indication with the voice, the identified input audio is broken down into smaller samples than they are transformed into single-channel images, known as spectrograms [25], once this process is completed, the data is sent to the neural network and the word spoken by the user is inferred. For the interpretation and representation of the input audios, once processed, language models are established, these models define how the audio is processed, what elements are involved and in what order [26]. In EULER a language model is defined for simple mathematical symbols and another for compound mathematical symbols.

1) Language model for simple symbols: The simple symbols were categorized in numerical values and in operators, this classification is proper. Operators are interpreted and represented symbolically, as shown in the examples in Table II.

TABLE II
OPERATORS REPRESENTATION

Verbal input	Representation
plus	+
alpha	α
empty set	$\emptyset$
angle	$\measuredangle$
cosine	cos

For the representation of numbers two cases can occur. The first case is when the number is only one digit, it is represented symbolically, similar to the operators. The other case is when the number is two digits or more, for this a syntactic analysis is performed to represent the number. In Figure 2, the process that is executed to represent the number is shown and Figure 3 shows the syntactic analysis performed to obtain the numerical value.

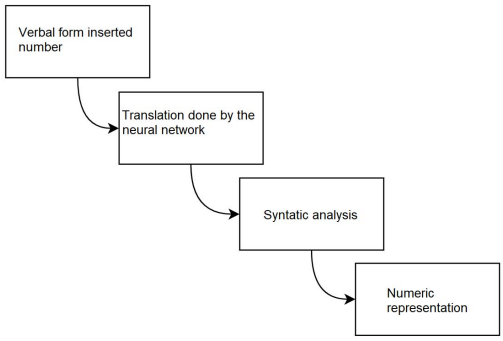


Fig. 2. Numerical value representation process

*This algorithm was developed for spanish language.

The parser takes the translation performed by the neural network and exchanges the equivalent numerical values, then concatenates them according to the positioning in base blocks 10. Considering the above, in EULER each value is added sequentially and visually displayed on the screen, also gives the auditory confirmation for the blind or low vision user.

2) Language model for compound symbols: The interaction with each compound symbol varies to avoid ambiguities, for example, if the user enters X raised to the n plus one, the editor could represent x^{n+1} or $x^n + 1$. To avoid this, a command-based protocol was chosen, in which the user interacts with the editor. In Table III, the interaction with the editor for the entry of composite symbols is exemplified.

The interaction is chosen in order to keep the user located and do not miss or make mistakes when you want to enter a

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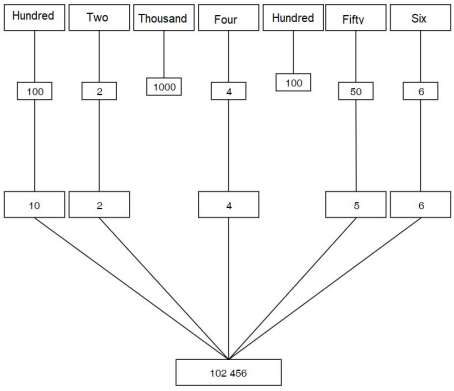


Fig. 3. Parsing to get the numerical value*

TABLE III
USERS INTERACTIONS EXAMPLES

Interactions		
Voice input (symbol)	fraction	power
Euler auditory output	Say the numerator	Say the base
Voice input (user response)	minus b plus a	two
Euler auditory output	say the denominator	say the exponent
Voice input (user response)	two a	a plus b
Euler auditory output	The expression Fraction *minus b plus a divided by two a*, end of fraction will be entered	The expression will be entered: *exponent two raised to a plus b, end of exponent*
Euler visual representation	$\frac{-b+a}{2a}$	2^{a+b}

mathematical expression, in addition, possible ambiguities are under control.

IV. EVALUATION AND RESULTS

With the data of the volunteers the training was carried out. For the evaluation of voice input, interpretation and representation, there have been three people, two with low vision and one with blindness.

A. Evaluations

For the tests carried out on the current development of voice input in EULER, the following have been taken into account:

- Numbers
- Basic operators (addition, subtraction, multiplication and division)
- Composite symbols (fraction and exponent)

The steps implemented to test the voice inputs with the elements mentioned above, were performed as follows.

- Users are told what commands they can use.
- The user is presented with an exercise that involves the commands provided.
- The user is assisted in case of doubt.
- The record of the results obtained with respect to the commands used by the user is kept.

B. Results

The incorporation of voice input functionality in EULER is still under development and testing, therefore, the results are in terms of changes and improvements. With the tests applied to the target users (blind and visually impaired), it was obtained that:

- The interpretation of the numbers was not always correct.
- Some symbols because of their similarity to numerical values are confused, for example, "*%*" represents it as "* 100". The tests were applied in Spanish, for this reason it is this result. This is because the symbol *"%"* is pronounced in Spanish "*por ciento*" and the saying "* *100*" is pronounced "*por cien*."
- It depends on how the value or symbol is pronounced, represents a symbol or number that does not correspond, for example, the users pronounced "uuuuno"[1], EULER represents "11".

From that findings, in development the following must be taken into account.

- It is necessary to improve the accuracy of numbers, for this you need to use another dataset with numbers.
- Initially a training method was used for word inferences, however, due to precision problems, a phoneme[2] training was chosen, with the aim of improving the proximity of the input audios.

For the other commands to be added to the voice input system, the results obtained with the target users must be taken into account and identify what new actions should be considered for the next tests.

V. CONCLUSIONS AND FUTURE WORKS

The learning and development of mathematics for people with visual disabilities is complicated because it is very visual. The representation of mathematical expressions through voice recognition is an alternative for the development of accessible mathematics.

EULER, which is a tool focused on blind and low vision people, with a function of entering expressions by voice facilitates the process of editing mathematics, offering an accessible and autonomous environment.

The change from training by inferences of words to training by phonemes, allows to treat the problem of proximity in the symbols or numbers, and allow the target user to have no problems of incorrect representation of symbols or numbers.

In the future it is planned to integrate more composite symbols which correspond to advanced levels in secondary and university education, such as systems of equations, integrals

[1]Pronunciation in Spanish for the number 1

[2]Defined as the phonological unit that cannot be broken down into successive smaller units and that is able to distinguish meanings, for example, the number "14" is associated with the word "fourteen", and this in turn is associated with phoneme / f / / o / / u / / r / / t / / e / / e / / n /.

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and propositional logic, in addition to the integration of more voice commands for greater application control, to navigate between menus and to switch between editing modes. Also develop a voice recognition version for the web version of the mathematical editor.

Another important project that could be addressed is to include support for multiple languages because there is currently support only for Spanish. To achieve this task, it is planned to look for people who are willing to voluntarily collaborate to record their voices and create the different *datasets* for different languages, it is also planned to use the databases offered by Google [27], since they contain a variety of audios compiled by the Google team such as music, speech, noise, among others.

It is necessary to expand the number of blind and low vision people to have more feedback on the functionality. Similarly, perform repeated tests with these people, to adjust to approach compliance to make mathematics accessible. Given the wide use of voice command functionalities, in the future, other types of users could be involved, for example, people with motor disabilities or elementary school children who have not yet developed the ability to write.

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AUTHORS AND AFFILIATIONS

Tribeth Rivas-Pérez is a Computer Science Engineering student at Technological Institue of Costa Rica, with experience in software development oriented to users with visual impairments. He is currently working Inclutec projects since 2018.

Abraham Meza-Vega is a Computer Science Engineering student at Technological Institue of Costa Rica, with experience in software development oriented to users with visual impairments. He is currently working Inclutec projects since 2018.

Verónica Mora-Lezcano is a Computer Science Engineer, with experience in software development oriented to users with visual impairments. She is currently working Inclutec projects since 2018. She received a degree in Computer Science Engineering in 2018, from the Technological Institute of Costa Rica.

Dionisio Palacio-Amador is a Information Technology Administration student at Technological Institue of Costa Rica, with experience in software development oriented to users with visual impairments. He is currently working Inclutec projects since 2017.

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Mario Chacón-Rivas is a Computer Engineer and a Doctor in Computer Applications. He is general coordinator of Inclutec and teacher at the Computer School of the Technological Institute of Costa Rica. He received the degree of computer engineer in 1993. He also received a master's degree in computer science in 1999, both degrees from the Technological Institute of Costa Rica. In 2016 he received the degree of Doctor of Computing from the University of Alicante, Spain.

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